



Name: Rhona Eloisa Lumbes

Role: Member

- **Specific Tasks performed**

The task that was assigned to me was to design and build the console-based client interface using Python. I did the overall structure of the interface, including the welcome screen, the login and registration of the accounts, and the admin dashboard. I implemented the client-side logic needed to connect to the backend API to send requests to the various endpoints, such as `login.php`, `register.php`, `add_to_inventory.php`, etc., which allows users to perform key functions such as viewing, adding, updating, and deleting inventory items. I also managed the user flow by creating a clear menu system and ensured that data received from the backend was presented in a clean and readable table format.

- **Code Contribution**

I created a modular function called `handle_api_response` to process server responses and convert HTTP status codes into understandable messages for users. I used the Python `requests` library to handle different HTTP methods such as `GET`, `POST`, `PUT`, and `DELETE` for interacting with the API. I also improved the visual design of the CLI by adding ANSI colors to make the interface more engaging. To avoid errors and improve formatting, I used raw strings and f-strings, especially for displaying structured text like tables and ASCII text art.

- **Challenges Encountered**

The main challenge that I always encountered while doing my part was the frequent 400 Bad Request errors, which were caused by the mismatched keys in the Python code I did and the PHP backend. I noticed that even small differences in naming led to the failed requests. Another challenge was formatting the table display in the terminal, where I had difficulties in aligning the text length and borders that made the output look unorganized. I also encountered some issues when combining code changes with my teammates, which required careful handling to avoid conflicts.



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- **Contribution in solving the problem**

To solve these issues, I carefully reviewed the PHP backend code to make sure that all JSON keys in my Python requests matched exactly with the expected variables. This helped eliminate the connection errors. For the UI problem, I applied fixed column widths and proper alignment techniques, such as right-justifying numbers, to maintain a consistent table layout. I also ensured smooth integration of code changes by organizing and merging updates properly, resulting in a stable and functional system.